Logistic Regression (Binary Classification)

Logistic Regression is used when the output variable is categorical, usually two classes.
Even though the name says regression, it is actually a classification algorithm.
Instead of predicting a numeric value like linear regression, logistic regression predicts a probability between 0 and 1.

Sigmoid Function (Logistic Function)
Logistic regression uses the **sigmoid function** to convert any value into a probability between **0 and 1**.

Threshold:
After getting the probability, we apply a decision threshold.

- Probability **< 0.6 → Spam**
- Probability **≥ 0.6 → Not Spam**

Multinomial Logistic Regression:
Binary logistic regression works only for two classes.
As well as Multinomial Logistic works on multiple classes.
Instead of one probability, the model predicts probabilities for each class.
EX : Cat, Dog, Monkey

Softmax Function
For multi-class classification, logistic regression uses the Softmax function.
Softmax converts scores into probabilities that sum to 1.

Confusion Matrix:
A Confusion Matrix is used to evaluate the performance of a classification model (like Logistic Regression).
It compares actual values vs predicted values.

1.True Positive (TP)
Actual = Fraud
Predicted = Fraud
The model correctly identifies a fraudulent claim.

2. True Negative (TN)
Actual = Legitimate Claim
Predicted = Legitimate
The model correctly identifies a genuine claim.

3. False Positive (FP)
Actual = Legitimate Claim
Predicted = Fraud
The model incorrectly flags a real claim as fraud.

4. False Negative (FN)
Actual = Fraud
Predicted = Legitimate
The model misses a fraudulent claim.

True Positive Correctly predicted positive

True Negative Correctly predicted negative

False Positive Predicted positive but actually negative

False Negative Predicted negative but actually positive

Accuracy (Classification Metric)
Accuracy measures how many predictions the model got correct out of all predictions.
It counts:

- **True Positives (TP)** → correctly predicted positive cases
- **True Negatives (TN)** → correctly predicted negative cases

Accuracy Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Recall (Sensitivity)
Out of all actual positive cases, how many did the model correctly identify.
How good the model is at finding the positives.
Recall=TP/(TP+FN)

Precision
Precision measures how accurate the model's positive predictions are.
Out of all the predictions the model labeled as positive, how many were actually positive?
Precision=TP/(TP+FP)

Precision–Recall Trade-off
In many ML models, precision and recall move in opposite directions when you change the classification threshold.
F1 Score
F1 Score combines Precision and Recall into a single metric.

ROC Curve (Receiver Operating Characteristic)
The ROC curve evaluates classification models across different thresholds.
The ROC Curve is a performance evaluation graph for classification models, especially binary classification problems.
It shows how well a model can separate two classes (for example: Fraud vs Not Fraud or Spam vs Not Spam).
True Positive Rate (TPR)

Also called **Recall / Sensitivity**
Out of all actual positive cases, how many did the model correctly predict?
False Positive Rate (FPR)
Out of all actual negative cases, how many did the model wrongly predict as positive?

AUC (Area Under Curve)
AUC (Area Under the Curve) is a single numeric metric that summarizes the performance of a classification model using the ROC curve.

- The ROC curve plots TPR (True Positive Rate) vs FPR (False Positive Rate).
- AUC measures the total area under that ROC curve.
- Probability that the model ranks a random positive example higher than a random negative example.
- Instead of checking model accuracy at one threshold, AUC evaluates performance across all possible thresholds.

1. Multiclass Classification
The model chooses one class from more than two possible classes.
Each observation belongs to only one category.

2. Multilabel Classification
One data point can belong to multiple classes at the same time.